

Lecture 27

Quadratic Form and Positive Definite Matrix

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Outline and Review

5 mins

Roadmap of MATH2041

Segment 1

Solving linear systems (Lec 3-15);
Solving least squares problem (Lect 16,17)

Segment 2

[Lec 18-20]: Two relatively independent parts:
—Determinant. [Important tool!]
—Linear transformation. [Advanced math perspective of matrix]

Segment 3

Lec 22-27: Eigenvalues and related.
Schedule:
—Eigenvalues. Lec 22-26
—Singular values. Lec 26-27
—Positive definite matrix and quadratic form. Lec 28.

Outline

Today: quadratic forms and definite matrices.

1) Positive (semi)-definite matrices

2) Application: conic [二次曲线]

After the lecture, you should be able to

1. Describe and prove the properties of positive definite and semi-definite matrices
2. Judge the type of curves given by quadratic equations

Three Decompositions

	Diagonalization	EVD	Full SVD
Form	$A = SDS^{-1}$	$A = VDV^T$	$A = U\Sigma V^T$
What matrix?	Some square matrices	Symmetric matrices	Any rectangular matrix
Middle component	Diagonal	Diagonal	Rectangular diagonal
Left & Right component	Invertible matrix	Orthogonal matrix	Orthogonal matrix
Vector form	N/A	$\sum_{j=1}^n \lambda_j \mathbf{v}_j \mathbf{v}_j^T$	$\sum_{j=1}^r \sigma_j \mathbf{u}_j \mathbf{v}_j^T$

Spectral Theorem

Theorem 23.1 [Spectral Theorem]

Any real symmetric matrix A can be written as

$$A = VDV^T = \sum_{j=1}^n \lambda_j \mathbf{v}_j \mathbf{v}_j^T. \quad (*)$$

where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ is a real diagonal matrix,

$V = [\mathbf{v}_1, \dots, \mathbf{v}_n]$ is a real orthogonal matrix.

(*): Eigenvalue decomposition (EVD) or eigendecomposition of A .

Property 23.1: All eigenvalues are real.

Property 23.2: Eigenvectors can form an orthonormal basis of $\mathbb{R}^n$.

(23.2a) Eigenvectors can form a basis of $\mathbb{R}^n$.

(23.2b) Can pick an eigenbasis that is orthonormal set.

Part I Positive/Negative Definite Matrices

Part I.1 Definition for Positive/Negative Definite Matrices

20 mins

Definition

Definition 27.1 (Quadratic form)

For any real symmetric matrix A , we say $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ is a quadratic form [二次形式].

Examples:

$$1. A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, f(x) = x^T A x = \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 2x_1^2 + 3x_2^2$$

$$2. A = \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix}, f(x) = x_1^2 + 4x_2^2 - 4x_1x_2$$

$$\begin{aligned} \begin{bmatrix} x_1 & x_2 \end{bmatrix} \left(\begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} x_1 - 2x_2 \\ -2x_1 + 4x_2 \end{bmatrix} \\ &= x_1(x_1 - 2x_2) + x_2(-2x_1 + 4x_2) \\ &= x_1^2 - 2x_1x_2 - 2x_2x_1 + 4x_2^2 \\ &= x_1^2 - 4x_1x_2 + 4x_2^2. \end{aligned}$$

$$3. A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}, f(x) = x_1^2 + 2x_2^2 + x_3^2 + 2x_1x_2 + 2x_2x_3.$$

Diagram showing the expansion of the quadratic form for matrix A:

- x_1^2 (from $1 \cdot x_1^2$)
- $2x_2^2$ (from $2 \cdot x_2^2$)
- x_3^2 (from $1 \cdot x_3^2$)
- $2x_1x_2$ (from $2 \cdot x_1x_2$)
- $2x_2x_3$ (from $2 \cdot x_2x_3$)

Reverse Direction

Example 1. $f(x) = 2x_1^2 - 7x_1x_2 + 8x_2^2$ Write it as quadratic form.

$$f(x) = x^T A x, \quad A = \begin{bmatrix} 2 & -7/2 \\ -7/2 & 8 \end{bmatrix}.$$

Not $A_t = \begin{bmatrix} 2 & t \\ -7-t & 8 \end{bmatrix}, \forall t \in \mathbb{R}.$

$$\begin{aligned} x^T A_t x &= x_1^2 + (t - 7 - t) x_1 x_2 + 8x_2^2 \\ &= x_1^2 - 7x_1 x_2 + 8x_2^2 = x^T A x. \end{aligned}$$

but A_t is NOT symmetric.

So don't use $\begin{bmatrix} 2 & -3 \\ -4 & 8 \end{bmatrix}$ or $\begin{bmatrix} 2 & -1 \\ -6 & 8 \end{bmatrix}$ for this quadratic form.

Remark

Remark: We have used quadratic form in a few earlier proofs.

$$\text{e.g. } B^T Bx = 0 \implies x^T B^T Bx = 0 \implies \| Bx \|^2 = 0 \implies Bx = 0.$$

Quadratic form

This relation is used in the proof of $\mathcal{N}(B^T B) = \mathcal{N}(B)$.

Definitions

$$\text{not } A \geq 0$$

$$\text{but } A \geq 0 \quad \geq 0$$

Definition 27.2 (definite matrices I)

A real symmetric matrix A is

- i) positive semidefinite (PSD) iff $\mathbf{x}^T A \mathbf{x} \geq 0, \forall \mathbf{x} \in \mathbb{R}^n$, denoted as $A \geq 0$;
- ii) positive definite (PD) iff $\mathbf{x}^T A \mathbf{x} > 0, \forall \mathbf{x} \in \mathbb{R}^n \setminus \{0\}$, denoted as $A > 0$;
- iii) indefinite if $\exists \mathbf{x}, \mathbf{y} \in \mathbb{R}^{n \times 1}$ s.t. $\mathbf{x}^T A \mathbf{x} < 0 < \mathbf{y}^T A \mathbf{y}$.

Simple case: diagonal matrix. $D = \text{diag}(\alpha_1, \alpha_2, \dots, \alpha_n)$.

$$x^T D x = (x_1 \quad x_2 \quad \cdots \quad x_n) \begin{pmatrix} \alpha_1 & & \\ & \ddots & \\ & & \alpha_n \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \alpha_1 x_1^2 + \alpha_2 x_2^2 + \cdots + \alpha_n x_n^2.$$

$$x^T D x \geq 0, \forall x \iff \alpha_1 \geq 0, \alpha_2 \geq 0, \dots, \alpha_n \geq 0.$$

$$x^T D x > 0, \forall x \neq 0 \iff \alpha_1 > 0, \alpha_2 > 0, \dots, \alpha_n > 0.$$

$$\exists x, y \in \mathbb{R}^n \text{ s.t. } x^T D x < 0 < y^T D y \iff \text{some } \alpha_i > 0 \text{ and some } \alpha_j < 0.$$

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Claim. Let $D = \text{diag}(\alpha_1, \dots, \alpha_n)$. Then

$$x^T D x \geq 0, \quad \forall x \in \mathbb{R}^n \quad \Longleftrightarrow \quad \alpha_1, \dots, \alpha_n \geq 0.$$

Proof. Since

$$x^T D x = \underbrace{\alpha_1}_{\geq 0} \underbrace{x_1^2}_{\geq 0} + \cdots + \underbrace{\alpha_n}_{\geq 0} \underbrace{x_n^2}_{\geq 0},$$

if $\alpha_1, \dots, \alpha_n \geq 0$, then clearly $x^T D x \geq 0$ for all x .

Conversely, assume $x^T D x \geq 0$ for all x .

For each j , choose $x = e_j$. Then

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}^T \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$0 \leq e_j^T D e_j = \alpha_j.$$

Thus $\alpha_j \geq 0$ for every j , so $\alpha_1, \dots, \alpha_n \geq 0$.

Definitions

Definition 27.3 (definite matrices II)

A real symmetric matrix A is

- i) negative semidefinite (NSD) iff $\mathbf{x}^T A \mathbf{x} \leq 0, \forall \mathbf{x} \in \mathbb{R}^n$, denoted as $A \leq 0$;
- ii) negative definite (ND) iff $\mathbf{x}^T A \mathbf{x} < 0, \forall \mathbf{x} \in \mathbb{R}^n \setminus \{0\}$, denoted as $A < 0$.

$\mathbb{R}^n$

Simple case: diagonal matrix.

$$x^T D x = \alpha_1 x_1^2 + \cdots + \alpha_n x_n^2.$$

$$x^T D x \leq 0, \forall x \iff \alpha_1 \leq 0, \dots, \alpha_n \leq 0.$$

$$x^T D x < 0, \forall x \neq 0 \iff \alpha_1 < 0, \dots, \alpha_n < 0.$$

Examples of Five Types of Diagonal Matrices

$$\text{PD: } \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

$$\text{PSD: } \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

$$\text{Indefinite: } \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}, \begin{bmatrix} -3 & 0 \\ 0 & +7 \end{bmatrix}$$

$$\text{ND: } \begin{bmatrix} -2 & 0 \\ 0 & -5 \end{bmatrix}$$

$$\text{NSD: } \begin{bmatrix} -2 & 0 \\ 0 & 0 \end{bmatrix} \quad \begin{bmatrix} -2 & 0 \\ 0 & -5 \end{bmatrix}$$

Part I.2 Checking Positive/Negative Definite Matrices

10 mins

Checking PSD/PD by Entries

For diagonal matrices,
PSD/PD/... can be judged by signs of entries.

For general matrices, is it the same?

Judgement: A is PD iff $a_{ij} > 0, \forall i, j$.

Checking PSD/PD by Entries

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PSD/PD/... can be judged by signs of entries.

For general matrices, is it the same.

Judgement: A is PD iff $a_{ij} > 0$.



Counter-example: $A = \begin{bmatrix} 1 & 5 \\ 5 & 2 \end{bmatrix}$.

$$x^T A x = x_1^2 + 2x_2^2 + 10x_1x_2.$$

Let $x_1 = 1, x_2 = -1$,

$$\text{then } x^T A x = 1 + 2 \times 1 - 10 \times 1 = 1 + 2 - 10 < 0.$$

Checking PD or PSD

If using the definition of PD/PSD to check,
it seems that we need to check sth holds for all x .

It seems impossible to check sth for all x ,
since there are infinitely many x .

$$x^T A x \geq 0, \\ \forall x \in \mathbb{R}^n$$

Fortunately, there is a simpler way to judge PD/PSD,
by just checking finitely many quantities.

Guess: What quantities?

Equivalent Conditions of PD and PSD

Theorem 27.1 [PD Equivalent Conditions]

For an arbitrary real symmetric matrix A , the following are equivalent:

- a) A is positive definite;
- b) All eigenvalues of A are **positive**;
- c) A can be written as $A = R^T R$ for some **invertible** matrix R .

Equivalent Conditions of PD and PSD

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Theorem 27.2 [PSD Equivalent Conditions]

For an arbitrary real symmetric matrix A , the following are equivalent:

- a) A is positive semidefinite;
- b) All eigenvalues of A are **non-negative**;
- c) A can be written as $A = R^T R$ for some **square** matrix R .

Remark: A is negative definite iff eigenvalues < 0 .
 A is indefinite iff some eigenvalues > 0 and some eigenvalues < 0 .

Interpretation of Theorem 28.1

Matrix is a “generalization” of number.

- Some concepts of numbers **can** be generalized to matrices.
e.g. Sum, product, and operation rules (associative, distributive)

e.g. Inverse

- Some concepts **cannot** be generalized.

e.g. Commutative rule of product

--“Non-negative” and “square” are classical concepts of numbers.

(in primary school, often learned right after sum/product of numbers)

Property-P: $x \geq 0$ iff $x=a^2$ for some $a \in \mathbb{R}$.

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Property-P: $x \geq 0$ iff $x=a^2$ for some $a \in \mathbb{R}$.

--Q: How to generalize “non-negative” to matrix?

Possibility 1: All entries are nonnegative.

Possibility 2: All eigenvalues are nonnegative (and symmetric).

-- **Thm 27.2 says:** Using Possibility 2, Property-P can be generalized to matrix.

Part I.2 Proofs

25 mins

Proof of Thm 28.1 (a) $\iff$ (b)

Suppose A is real symmetric.

Prove:

All eigenvalues of A are positive

iff

A is positive definite.

Proof of Thm 28.1 (a) $\Leftrightarrow$ (b)

(b) $\Rightarrow$ (a) If all eigenvalues are positive, and A is real symmetric, then A is positive definite.

Hint: Use EVD.

Need to prove $x^T A x > 0$, $\forall x \in \mathbb{R}^n \setminus \{0\}$.

By EVD, $A = \sum_{j=1}^n \lambda_j v_j v_j^T = V D V^T$, $\lambda_j > 0$, $\forall j$.

Method 1. Let $y = V^T x$. Then

$$x^T A x = x^T (V D V^T) x = \underbrace{(x^T V)}_{y^T} \cdot D \cdot \underbrace{(V^T x)}_y$$

$\begin{matrix} x^T V & V^T x \\ & \searrow \quad \nearrow \\ & y^T & y \end{matrix}$

$$\left(D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \right) = \underbrace{y^T D y}_{\substack{\text{quad-form, diagonal } D}} = \sum_{j=1}^n \underbrace{\lambda_j}_{>0} \underbrace{y_j^2}_{\geq 0} > 0.$$

Need $y \neq 0$: otherwise, if $y = \underbrace{V^T}_{\text{invertible}} x = 0$, then $x = \underbrace{V}_{\text{invertible}} \cdot y = 0$, contradicting $x \neq 0$.

(invertible matrix) $\cdot$ (non-zero vec) = non-zero vec.

Proof of Thm 28.1 (a) $\Leftrightarrow$ (b)

(b) $\Rightarrow$ (a) If all eigenvalues are positive, and A is real symmetric, then A is positive definite.

Method 2:

$$x^T A x = x^T \left(\sum_{j=1}^n \lambda_j v_j v_j^T \right) x$$

$$= \sum_{j=1}^n \lambda_j \boxed{x^T v_j v_j^T x}$$

dim-checking

$$\left(x^T V \right) D \left(V^T x \right) \leftarrow = \sum_{j=1}^n \lambda_j \underbrace{\left(\underbrace{x^T v_j}_{1 \times n} \right) \left(\underbrace{v_j^T x}_{n \times 1} \right)}_{|x|^2}$$

$x^T v_j = v_j^T x \in \mathbb{R}, a^T b = b^T a.$

$y_j^T y_j, y^T y, y^2, y,$
 $y_j y_j^T$ "X"

$$y^T D y \leftarrow = \sum_{j=1}^n \lambda_j \boxed{(x^T v_j)^2} \geq 0.$$

or y_j^2 , where $y_j = x^T v_j \in \mathbb{R}$

Since $\lambda_j > 0$ for all j , equality holds only when $\boxed{x^T v_j} = \boxed{0}$ for all j . $\rightarrow V^T x = 0$

Because $v_1, \dots, v_n$ form an **basis**, this implies $x = 0$.

$x^T z = 0, \forall z \in \mathbb{R} \rightarrow$

$\Downarrow$
 $x = 0$

Therefore, for every $x \neq 0$,

$$x^T A x > 0.$$

Proof of Thm 28.1 (a) $\Leftrightarrow$ (b)

(a) $\Rightarrow$ (b) If A is positive definite, then all eigenvalues are positive.

Hint: Use EVD.

Suppose $A = VDV^T = \sum_{j=1}^n \lambda_j v_j v_j^T$ and $x^T A x > 0, \forall x \neq 0$.

Need to show $\lambda_k > 0, \forall k$.

$$x = e_j, \quad e_j^T D e_j = \alpha_j.$$

Idea. Pick a smart x .

Pick $x = v_k$. Then

$$0 < x^T A x = v_k^T \left(\sum_{j=1}^n \lambda_j v_j v_j^T \right) v_k = \lambda_k \underbrace{(v_k^T v_k)}_{=1} \underbrace{(v_k^T v_k)}_{=1} + \sum_{j \neq k} \lambda_j \underbrace{(v_k^T v_j)}_{=0} \underbrace{(v_j^T v_k)}_{=0} = \lambda_k.$$

Therefore, $\lambda_k > 0$.

Since k is arbitrary, $\lambda_k > 0, \forall k$.

Proof of Thm 28.1 (a) $\iff$ (b)

Suppose A is real symmetric.

Prove:

A is positive definite

iff

$A = R^T R$ for some invertible matrix R .

Proof of Thm 28.1 (a) $\Leftrightarrow$ (c).

(c) $\Rightarrow$ (a) If $A = R^T R$ for some invertible matrix R , then A is PD.

Proof. Suppose $A = R^T R$. If $x \neq 0$, then $Rx \neq 0$
(otherwise, $x = \underline{R^{-1}0} = 0$).

Then

$$x^T A x = x^T R^T R x = (Rx)^T (Rx) = \|Rx\|^2 > 0.$$

So A is PD.

This proof is similar to the proof we've seen before:

$$x^T B^T B x = 0 \Rightarrow \|Bx\|^2 = 0 \Rightarrow Bx = 0$$

Proof of Thm 28.1 (a) $\Leftrightarrow$ (c).

(a) $\Rightarrow$ (c) If A is PD, then $A = R^T R$ for some invertible matrix R .

Proof. A is PD $\Rightarrow$ eigenvalues > 0 . *need to construct R , based on A .*

Let $A = \underline{VDV^T}$, where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ and $\lambda_i > 0, \forall i$.

Let

$$D^{1/2} = \text{diag}(\lambda_1^{1/2}, \dots, \lambda_n^{1/2}), \quad R = D^{\frac{1}{2}} V^T.$$

$$R^T = V \cdot D^{\frac{1}{2}}.$$

Since $D^{1/2}$ and V^T are invertible, R is invertible.

Also,

$$R^T R = (VD^{\frac{1}{2}})(D^{\frac{1}{2}}V^T) = VDV^T = A.$$

Therefore, $A = R^T R$.

$$\underbrace{(VD^{\frac{1}{2}})}_{R^T} \underbrace{(D^{\frac{1}{2}}V^T)}_R$$

$$R^T = (D^{\frac{1}{2}}V^T)^T = V(D^{\frac{1}{2}})^T = V \cdot D^{\frac{1}{2}}$$

日期: /

$$A = V D V^T$$

$$= \underbrace{(V D^{\frac{1}{2}})}_{R^T} \underbrace{(D^{\frac{1}{2}} V^T)}_R$$

Remarks

We have shown proofs for Thm 28.1.

(PD)

The proof of Thm 28.2 is quite similar:
the difference is to change strict inequalities to inequalities.

Skip here; left as exercise.

>

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Property

Property 28.1: If A is positive definite, then A is invertible.

note. PSD matrix can possibly be non-invertible.

Proof method 1 (determinant and eigenvalues):

Since A is PD, all eigenvalues are > 0 : $\lambda_1, \dots, \lambda_n > 0$.

Therefore, $\det(A) = \lambda_1 \lambda_2 \dots \lambda_n > 0$. ($\neq 0$)

So A is invertible.

Proof method 2 (product of invertible matrices):

By EVD, $A = V D V^T$, where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$.

Since $\lambda_i > 0$ for all i , D is invertible.

Since V , D , and V^T are all invertible, their product $A = V D V^T$ is also invertible.

Remark: Terminology

In some textbooks, positive definite matrices must be symmetric.

In others, positive definite matrices can be non-symmetric,
(so you may see terms “symmetric PD”, “s.p.d matrix” or “symmetric PSD”)

Example: $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

Does it satisfy $\mathbf{x}^T A \mathbf{x} > 0, \forall \mathbf{x} \in \mathbb{R}^2 \setminus \{0\}$? Yes.

$$B = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}$$

$$\mathbf{x}^T A \mathbf{x} = \mathbf{x}^T B \mathbf{x} > 0, \forall \mathbf{x} \neq 0$$

Is it symmetric? No.

Some books still say “A is PD”; but in our class, we say “A is not PD”.

In MAT2041:

Positive definite matrices must be symmetric (by our definition)
(PSD matrix also must be symmetric)

Reading: Application in Optimization

Application 1: Convex analysis; convex optimization.

A twice-differentiable function f is convex iff $\nabla^2 f(x) \succeq 0, \forall x \in \mathbb{R}^n$.

Application 2: Optimization.

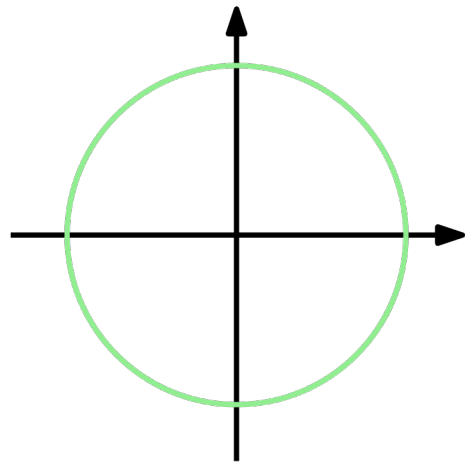
Necessary condition for optimal solution:

$$\nabla f(x^*) = 0 \text{ and } \nabla^2 f(x^*) \succeq 0.$$

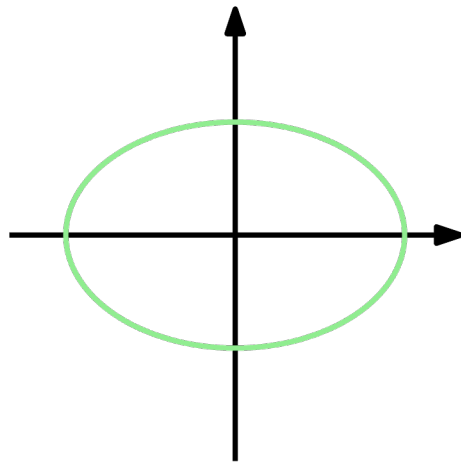
Part II Application: Curve Classification

10 mins

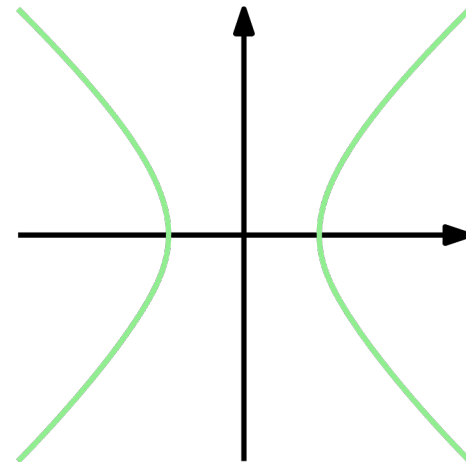
Curves



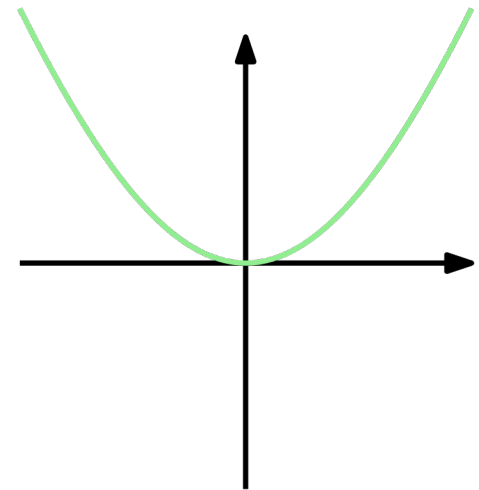
Circle



Ellipse



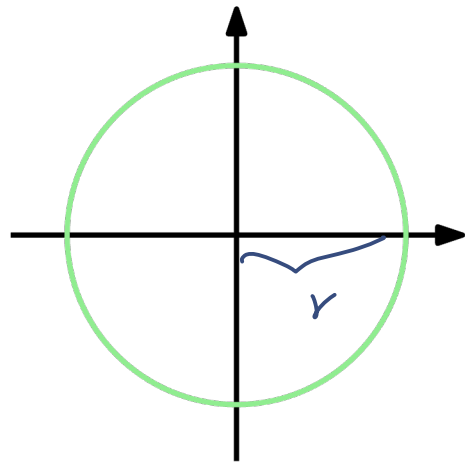
Hyperbola



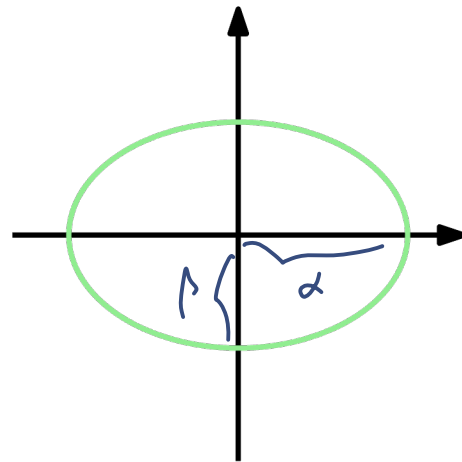
Parabola

Typical equations:

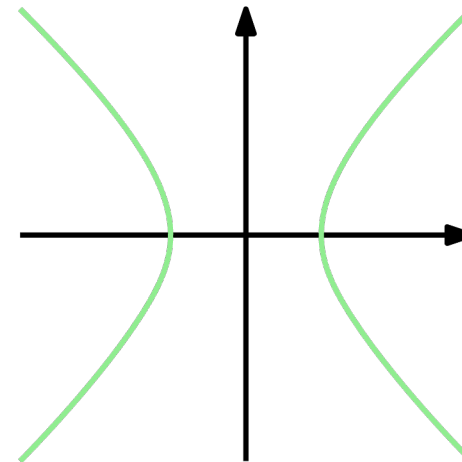
Curves



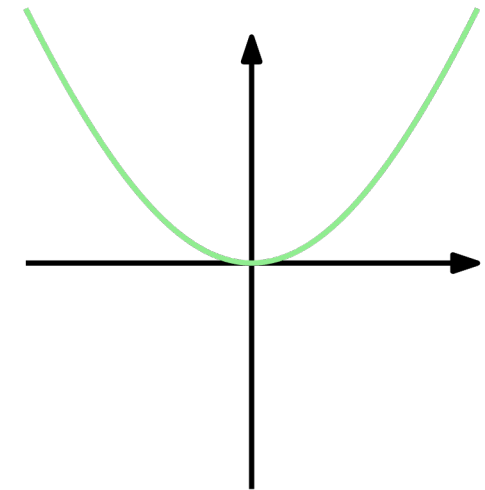
Circle



Ellipse



Hyperbola



Parabola

Typical equations:

$$x^2 + y^2 = r^2$$

Circle

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Ellipsoid

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = \pm 1$$

Hyperbola

$$x^2 = \alpha y \text{ or } y^2 = \alpha x$$

Parabola

Curves

$$\frac{x^2}{\alpha^2} + \frac{y^2}{\beta^2} = 1$$

Ellipsoid

$$[x, y] \begin{bmatrix} 1/\alpha^2 & 0 \\ 0 & 1/\beta^2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 1.$$

Diagonal entries: > 0

$$\frac{x^2}{\alpha^2} - \frac{y^2}{\beta^2} = \pm 1$$

Hyperbola

$$[x, y] \begin{bmatrix} 1/\alpha^2 & 0 \\ 0 & -1/\beta^2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 1.$$

Diagonal entries: $> \text{ and } < 0$

Determining Shape of Conic Sections: Trial

Example 27.1: What is the shape of the conic section of

$$5x_1^2 - \underline{6x_1x_2} + 5x_2^2 = 4?$$

We express it as a **matrix form**.

$$[x_1, x_2] \begin{bmatrix} 5 & -3 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 4 \quad (**)$$

$$x^T A x = c$$

Derivation of Criteria: EVD

If $A = VDV^T$, what to say about equation $\mathbf{x}^T A \mathbf{x} = c$?

$$\mathbf{x}^T A \mathbf{x} = c \iff \underbrace{(\mathbf{x}^T V)}_z D \underbrace{(V^T \mathbf{x})}_z = c \dots \text{Then?}$$

Let $(\mathbf{x}^T V) D (V^T \mathbf{x}) = \underbrace{z^T D z}_{\text{vec} * \text{diagonal matrix} * \text{vec}}$

vec * matrix * vec vec * diagonal matrix * vec

What Determine Shape?

$$\mathbf{x}^T A \mathbf{x} = c \quad (*)$$

$$A = V D V^T$$

entries of D are eig-vals of A

Eq. (*) becomes $\mathbf{z}^T \underline{D} \mathbf{z} = c \quad (**)$

A's eigenvalues

++



D: ++



(**) is ellipsoid

A's eigenvalues

+-



D: +, -



(**) is hyperbola

Algorithm: Transform to Standard Form

Algorithm 28.1:

Input: $\mathbf{x}^\top A \mathbf{x} = c \quad (*)$

Step 1 (EVD): $A = V D V^\top$

Step 2 Let $\mathbf{z} = V^\top \mathbf{x}$ Here, V is orthogonal matrix (P1)

Step 3 Eq. (*) becomes $\mathbf{z}^\top D \mathbf{z} = c$ Here, D is diagonal matrix (P2)
Or $\mathbf{z}^\top \frac{D}{c} \mathbf{z} = \mathbf{1} \quad (**)$

Step 4 Judge:

$(**)$ is shape-S (S in {ellipsoid, hyperbola, parabola}) This is easy, due to (P2)

Conclusion:

The original conic section $(*)$ is shape-S;
Its standard form is $(**)$.

Curves represented by $(**)$ and $(*)$
are equivalent [全等], when (P1) holds.

(Requires a proof; skip here;
See next page for more discussions)

Conic Section: Classification By Eigenvalues of A

Expression	Signs of eigenvalues of A	Simplified form (standard form)	Conic section (shape)
$\mathbf{x}^T A \mathbf{x} = \alpha$	$+, +$ or $-, -$	$\frac{z_1^2}{\alpha^2} + \frac{z_2^2}{\beta^2} = 1 \text{ or } -1$	Ellipsoid (Or empty set)
$\mathbf{x}^T A \mathbf{x} = \alpha$	$+, -$	$\frac{z_1^2}{\alpha^2} - \frac{z_2^2}{\beta^2} = 1 \text{ or } -1$	Hyperbola
...	contain 0	...	...

$\lambda_1, \lambda_2 > 0$,
 $\alpha > 0$:
 ellipsoid;
 $\lambda_1, \lambda_2 < 0$,
 $\alpha > 0$:
 empty set;
 etc.

Skip
this case

The case of containing 0 is skipped in our class,
due to time reason.

Reading: Identical Shape

We explain why judging by eigenvalues works.

Claim: Consider two point sets:

$$\Omega_1 = \{\mathbf{x}: \mathbf{x}^\top A \mathbf{x} = \alpha\} \quad \Omega_2 = \{\mathbf{z}: \mathbf{z}^\top B \mathbf{z} = \alpha\}$$

If there exists an **orthogonal matrix** V s.t. $B = V A V^\top$,

Then the two sets are “equivalent” in the following sense:

by combination of rotation and reflection of Ω_1 , we can make it to completely overlap with Ω_2 .

[in Chinese: 两个几何图形是全等的]

[This definition is also used in “identical triangles”; 全等三角形]

Why?

If V is an orthogonal matrix, then $\mathbf{y} = V \mathbf{x}$ is an **isometric transformation** [等距变换] that preserves distance, i.e. $\|V \mathbf{x} - V \mathbf{u}\| = \|\mathbf{x} - \mathbf{u}\|, \forall \mathbf{x}, \mathbf{u}$.

Property:

Isometric transformation keeps the shape of a geometrical subject unchanged.

Determining Shape of Conic Sections

Example 27.1: What is the shape of the conic section of $5x_1^2 - 6x_1x_2 + 5x_2^2 = 4$? Transform it to standard form.

$$\Leftrightarrow 8 - \left(\frac{1}{\sqrt{2}} (x_1 - x_2) \right)^2 + 2 - \left(\frac{1}{\sqrt{2}} (x_1 + x_2) \right)^2 = 4$$

Step 1 (EVD):

Eigenvalues: $\lambda_1 = 8, \lambda_2 = 2$.

Eigenvectors $V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

$$\begin{aligned} & V^T x \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} x_1 - x_2 \\ x_1 + x_2 \end{bmatrix} \end{aligned}$$

Step 2 (Change of variables):

$$z = V^T x = \frac{1}{\sqrt{2}} \begin{bmatrix} x_1 + x_2 \\ -x_1 + x_2 \end{bmatrix}$$

Step 3 (Transformed equation):

$$8z_1^2 + 2z_2^2 = 4 \Rightarrow \frac{z_1^2}{0.5} + \frac{z_2^2}{2} = 1$$

This is the transformed standard form.

Step 4 (Shape):

Since $\lambda_1, \lambda_2 > 0$, the conic is an ellipse.

日期: /

Another example: why eigenvalues?

Shape of curves $\xleftarrow{\text{determine}}$ $\begin{matrix} \lambda_1 \\ \lambda_2 \end{matrix}$

Math: helps reveal the essence of
the world.



Summary Today (Write Your Own)

One sentence summary:

Detailed summary:

Summary Today (Instructor)

One sentence summary:

We studied the quadratic form, PSD matrix and conic.

Detailed summary:

PSD matrices.

- Definition: A is PSD (resp. PD) iff $\mathbf{x}^\top A \mathbf{x} \geq$ (*resp.* $>$) $0, \forall \mathbf{x} \in \mathbb{R}^n \setminus \{0\}$.
- Definitions of NSD, ND and indefinite matrix.
- Equivalent properties: A is PSD (resp. PD)
 - All eigenvalues are non-negative (resp. positive).
 - $A = RR^\top$ for square (resp. invertible) matrix R .

Quadratic form and conic section.

- Quadratic form $f(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x}$.
- Write 2-var, deg-2 equation as $\mathbf{x}^\top A \mathbf{x} = c$
- Transform $\mathbf{x}^\top A \mathbf{x} = c$ to standard form:
 - let $A = VDV^\top$, then $\mathbf{y} = V^\top \mathbf{x}$, get $\mathbf{y}^\top A \mathbf{y} = c$.
- Judge shape by eigenvalues of A :
 - (+, +): ellipse, PD. (+, -): hyperbola, indefinite.

Final Remarks

This is the end of teaching new contents.

There is one more class on May 6 Wed;
but no new contents.

Likely a review session; specific form TBD.

Final Exam:

Lec 1-27.

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Appendix: Conic Section with Linear Terms

With Linear Terms

How to check shape of

$$2x^2 + 3xy - 5y^2 - 7x + 8 = 0?$$

What type of curve does a general equation represent?

$$ax^2 + bxy + cy^2 + dx + ey + f = 0$$

Challenging: cross term xy . 2-var, degree-2 (二元二次方程)

We express it as a **matrix form**:

$$\mathbf{u}^\top A \mathbf{u} + \mathbf{w}^\top \mathbf{u} + f = 0.$$

$$\text{where } \mathbf{u} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad A = \begin{bmatrix} \square & \square \\ \square & \square \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} \square \\ \square \end{bmatrix}$$

2-Var Quadratic Equation: Definitions

$$ax^2 + bxy + cy^2 + dx + ey + f = 0 \quad (27.1)$$

We express it as a **matrix form**:

$$\mathbf{u}^T A \mathbf{u} + \mathbf{w}^T \mathbf{u} + f = 0.$$

Where: $\mathbf{u} = \begin{bmatrix} x \\ y \end{bmatrix}$, $A = \begin{bmatrix} a & b/2 \\ b/2 & c \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} d \\ e \end{bmatrix}$

Definition: $f(\mathbf{u}) = \mathbf{u}^T A \mathbf{u}$ is called the **quadratic form** [二次形式] associated with the quadratic equation (27.1).

The graph corresponding to (27.1) is called the **conic section, or conic** [圆锥曲线].

Reading: Quadratic Equations With Linear Terms

For reading; will not test problems with linear terms in hw/exams.

Example 27.1b: What is the shape of the conic section of

$$4x_1^2 - 4x_1x_2 + 4x_2^2 + 4x_1 - 8x_2 = 6 \quad (\text{b.1})$$

Step 0&1&2: Same as before.

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = V^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} x_1 + x_2 \\ -x_1 + x_2 \end{bmatrix}; \quad \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = V \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} y_1 - y_2 \\ y_1 + y_2 \end{bmatrix}$$

Step 3: Rewrite equation.

$$(\text{b.1}) \text{ becomes } 2y_1^2 + 6y_2^2 + 2\sqrt{2}(y_1 - y_2) - 4\sqrt{2}(y_1 + y_2) = 6$$

$$\text{i.e. } 2y_1^2 + 6y_2^2 - 2\sqrt{2}y_1 - 6\sqrt{2}y_2 = 6, \quad \text{completing squares [配方法]}$$

$$\text{Or } 2(y_1 - 1/\sqrt{2})^2 + 6(y_2 - 1/\sqrt{2})^2 = 8.$$

$$\text{Or } \frac{(y_1 - 1/\sqrt{2})^2}{4} + \frac{(y_2 - 1/\sqrt{2})^2}{4/3} = 1. \quad (\text{b.2})$$

This represents an ellipse. So Eq. (b.1) also represent an ellipse.